TWO SIDES OF THE SAME COIN

Divisibility asks whether one integer divides another exactly. Modulo computes the evidence: the remainder answers the question.

MODULO AND DIVISIBILITY

$a \bmod n$

split by **whether the remainder is 0**

✓

CASE 1

$a \bmod n = 0$

n divides a

the remainder vanishes: n divides a with nothing left over

$$n \mid a$$

E.G. $24 \bmod 6 = 0$, so $6 \mid 24$

✗

CASE 2

$a \bmod n \neq 0$

n does not divide a

the remainder is positive and tells by how much the division misses

$$n \nmid a$$

E.G. $25 \bmod 6 = 1$, so $6 \nmid 25$

TAKEAWAY

Divisibility is the yes-or-no question; modulo is the tool that answers it and, when the answer is no, quantifies the shortfall.